Adaptive Stokesian Dynamics for interparticle interactions

MAT226C Final Project

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1. Adaptive Stokesian Dynamics

Stokesian Dynamics gives a simple relationship between the velocities and the forces of particles as F = RU. Resistance matrix, R is a 11N x 11N matrix, while force and velocity vectors are of size 11N. Velocities of individual particles are obtained given the forces acting on them, which is used to obtain particle coordinates using $\frac{dx}{dt} = u(t, x)$. It is relatively easy to simulate flow of a rigid aggregate where the near-field interactions can be ignored. Forward Euler, 2-step Adams-Bashforth and fixed-step RK4 methods can be used for timestepping and updating new particle coordinates using previous coordinates and particle velocities. However, when particles in an aggregate interact with each other through attractive and repulsive forces, near-field interactions are pivotal to capture the dynamics of the system and timestepping becomes crucial to prevent spheres from overlapping. Runge-Kutta-Fehlberg method is used to employ variable timestepping depending on minimum distance between particles.

1.1 Brief overview of Stokesian Dynamics

When the translational and rotational velocities on particles are known, the corresponding forces and torques acting can be determined by using the Grand Resistance Matrix, R. Similarly, when the forces and torques acting on particles are known, translational and rotational velocities can be determined.

$$\begin{pmatrix} F \\ L \\ S \end{pmatrix} = R \begin{pmatrix} U - u^{\infty} \\ \Omega \ - \Omega^{\infty} \\ -E^{\infty} \end{pmatrix}$$

U$-u^{\infty}$ $and\ \Omega - \Omega^{\infty}$ are vectors of length 3N for N particles. $-E^{\infty}$is a vector of length 9N. F and L are vectors of length 3N. S is the stresslet of length 9N and the resistance matrix is of size 11N x 11N.

Here, the far-field resistance matrix, R^{ff} is dense, depends on the configuration of particles and is obtained for all particle pairs using Faxen's laws. The near-field matrix, R^{nf} is sparse and accounts for all pairs within a critical radius r* from a particle centre.

Force vector incorporates external forces as well as interparticle interactions, and in this low Reynolds regime, forces are additive.

1.2 Interparticle interactions and parameters

van der Waals + Born repulsion

$$F_p = \frac{\left[-\frac{A}{12ah^2} + \frac{A}{360a}\left(\frac{\sigma}{a}\right)^6 \frac{1}{h^8}\right]}{6\pi\eta a U_s} \quad 7.2$$

where A is the Hamaker's constant,

a is the particle size, h(=h_s/a) is the non-dimensional surface-surface separation.

σ is the minimum distance at which the potential is zero.

η is the fluid viscosity and U_s is the steady state particle velocity.

The parameters used for both van der Waals and Born repulsion are given in table below.

Parameters	*Values*
A	1×10^{-20}J
a	10 nm
σ	0.2 nm
η	$10^{-5} kg/ms$
U_s	0.1 m/s

Table 1:Parameters chosen for the DLVO-type forces

Hamaker's constant is always in the range of 10^{-19} to 10^{-21} J. Minimum particle size should be 10 nm < a <100 nm to be in the colloids size range.

New positions are obtained by using

$$\mathrm{U} = \mathrm{u}_\infty + \mathrm{R}^{-1}[\mathrm{F_p} + \mathrm{F_e}] \quad 7.3$$

These derivations are shown in the appendix of this chapter.

2.1 Problem statement

The case studies here involves aggregates of N=2,12 and 96 particles of unit radius, where the particles are separated from each other by an interparticle distance, d =2.01. External force of unity acts on all particles in the z-direction, while the particles interact amongst themselves with attractive(vanderwaal's) and repulsive(Born) forces. Equilibrium surface-to-surface distance (minimum potential) is obtained by using $h_o = 0.5673\frac{\sigma}{a}$, which gives the interparticle distance of 2.011436 for the parameters used.

2.2 Implementation of variable timestepping RK

Bogacki-Shampine 3(2) and Runge-Kutta-Fehlberg[3] 4(5) were implemented with Stokesian Dynamics. Bogacki-Shampine is better suited compared to RK-Fehlberg since the former requires less function calls and faster compared to the latter. Function calls here refer to the matrix computations to obtain particle velocities at various steps using the forces calculated using particle positions and previous velocities. The time complexity of the matrix computations are $O(N^3)$ and hence a low-order variable timestepping RK works better for large systems. For rigid aggregate, hydrodynamic radius remains constant with time and for flexible aggregates, time-averaging is required once stability of interparticle interactions are ensured with less fluctuations in timestepping.

2.3 Bogacki-Shampine setup

Two estimates for the updated particle positions are calculated using B coefficients $\{2(3)\}$. The error between these two estimates calculates the accuracy of the step. A penalty is added to this error when interparticle distance between any two spheres goes less than 2.0, indicating overlap of particle. Maximum error$(atol = 9 \times 10^{-3}; rtol = 10^{-6})$ is obtained from sphere coordinates from the previous timestep. The step is accepted if the error is less than the maximum error and minimum interparticle distance is above 1.999. Tolerances for maximum error set up by trail-and-error by observing variations in error between estimates.

Timestep adjustment: Three cases for error ratio = error/maximum error

$$new\ timestep = factor\ \times error\ ratio^{\left(\frac{1}{p+1}\right)}$$

i) If step is rejected: Timestep adjustment factor of 0.8

ii) If step is accepted:

   a. Ratio < 0.6: Timestep is adjusted twice. One which accounts for the change in average interparticle distance and again with a timestep adjustment factor of 1.5. This is done to ensure the timestep doesn't become too small i.e. 10^{-6} or less

   b. Ratio < 1: Timestep adjustment factor of 1.5

Although, the parameters and factors chosen here work well with small aggregate sizes of N=12,21 and 96, the system might stay unstable for longer periods for large aggregates.

3. Implementation

3.1 Equilibrium distance for 2 particles

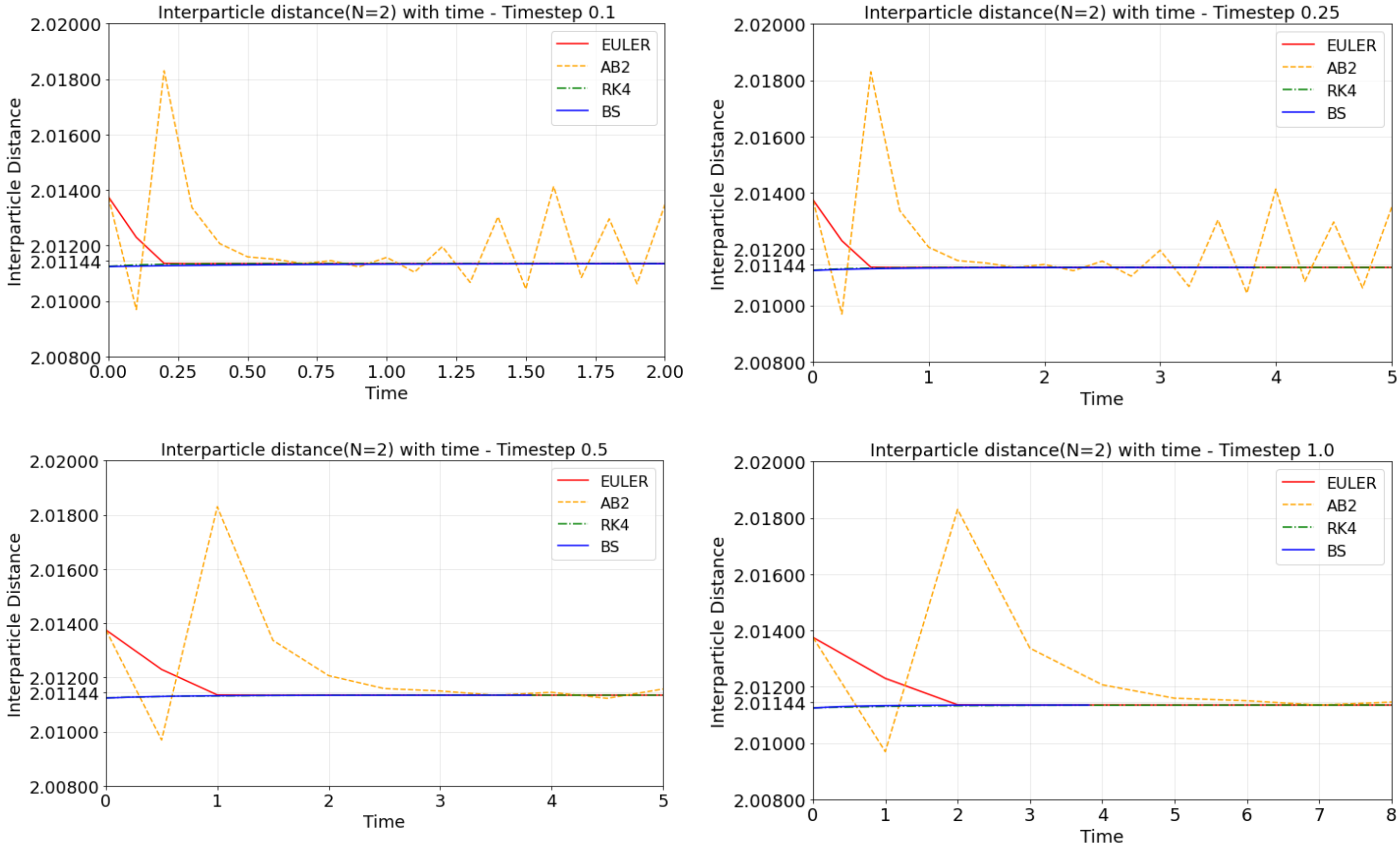

Figure 1: Two particles reaching an equilibrium distance for varying timesteps(dt=0.1[top left],dt=0.25[top right],dt=0.5[bottom left],dt=1[bottom right]) and solvers compared with the use of variable timestepping RK (Bogacki-Shampine)

For a simple system of 2 particles, variable timestepping RK was setup and its performance was compared with other solvers. It is relatively easy to obtain equilibrium for a small system and here RK4 and the variable timestepping RK perform better as the system stabilizes quickly. For fixed timestep methods such as Euler or AB2, small timesteps stabilize the system quickly as Euler requires minimum 2 steps and AB2 required minimum 6 steps to attain distances close to equilibrium. AB2 with fluctuations after reaching near-equilibrium distance shows how sensitive the system is.

3.2 Stability for system with N=12

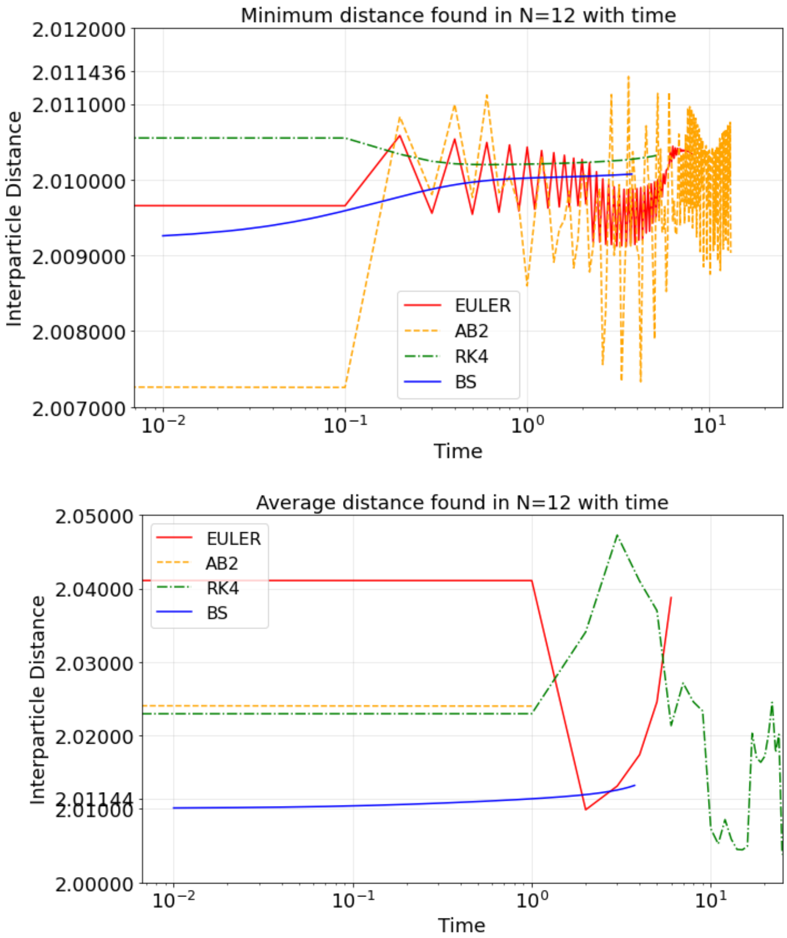

Figure 2: Minimum of interparticle distance(top) and average interparticle distance(bottom) observed in an aggregate of size N=12 using various methods

A timestep of 0.1 was used for Euler, AB2, RK4 and variable timestep RK and the minimum and average of distances between particles were observed as seen in *Figure 2*. Under ideal conditions, all the particles would end up close to equilibrium distance between each other which in this case would be 2.011436. Euler and AB2 being lower order methods are unstable with this timestep and the distance between particles are fluctuating. Interparticle distance keeps increasing slowly with variable timestepping RK, as they slowly increase with time due to repulsive forces. Although, this comparison isn't good enough provided the timesteps of variable timestep RK is ~0.0045, once the variations in errors slow down as seen in *Figure 3*.

Accepted timesteps and error between estimates are plotted against time for the Bogacki-Shampine method in Figure 3. The system reaches a pseudo-steady state after some time as there is only a small increase in timestep. Every aggregate configuration would have different timesteps required to yield a system where particles are not overlapping or exploding due to the attractive and repulsive forces. Bogacki-Shampine or higher order methods could be used to yield optimum timestep range at which the system stays relatively stable, and that timestep could be utilized by other lower order methods.

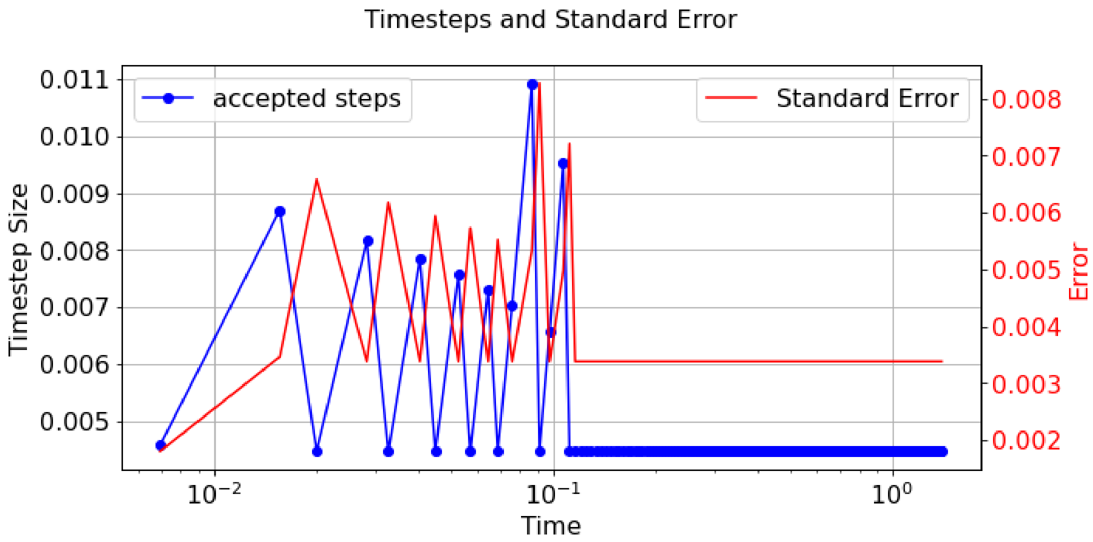

Figure 3: Varying timesteps based on error between estimates for N=12(Bogacki-Shampine)

3.3 Comparison across different particle sizes

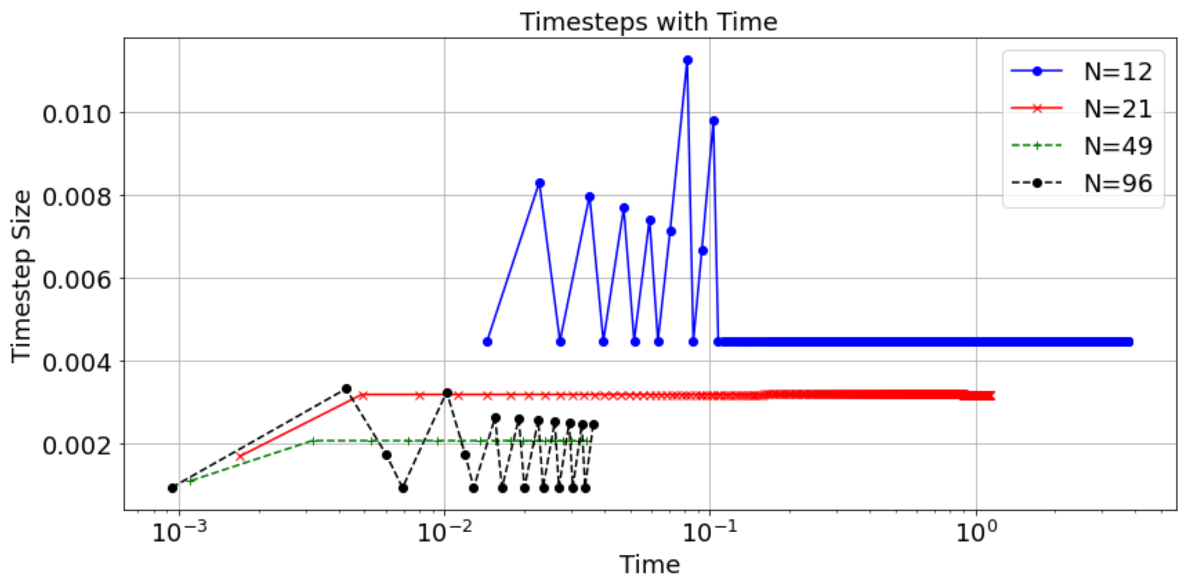

Figure 4: Varying timesteps based on error between estimates for N=12,21,49,96 (Bogacki-Shampine)

Change in timestep was observed for aggregates of four different sizes based on the error estimates using Bogacki-Shampine method. It was observed that as the particle size decreases, the timestep needed to stabilize the system becomes small(as expected). Figure 4 shows variation in timestep for two different types of aggregates, perfectly symmetrical (N=21 and N=49) and irregular (N=12 and N=96). Similar to N=12, the flucatuations in timesteps are expected to reduce before it reduce to an a almost steady timestep.

4. Conclusion

Variable timestepping RK has diminished the possibilities of overlapping particles, while the large-scale matrix computations makes it 3-5 times more time consuming. Instead, variable timestep Euler's method[4] could be more useful when it comes to large systems.

4. References

1) Brady, John F., and Georges Bossis. "Stokesian dynamics." Annual review of fluid mechanics 20.1 (1988): 111-157.
2) Townsend, A. K. (2018). Generating, from scratch, the near-field asymptotic forms of scalar resistance functions for two unequal rigid spheres in low-Reynolds-number flow. arXiv preprint arXiv:1802.08226. (https://github.com/Pecnut/stokesian-dynamics)
3) Runge-Kutta-Fehlberg method (https://paws.kettering.edu/~ktebeest/math305/rkf45.pdf)
4) C.W. Gear, Numerical Initial Value Problems in ODE's, and Prof. Feldman's notes (https://personal.math.ubc.ca/~feldman/math/vble.pdf)